MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (ACCURATE),

STYLE RB52

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3). USE MIL-R-39005/1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RB52, accurate, wire-wound, fixed resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-93 - Resistors, Fixed, Wire-wound (Accurate), General Specification for.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-93, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

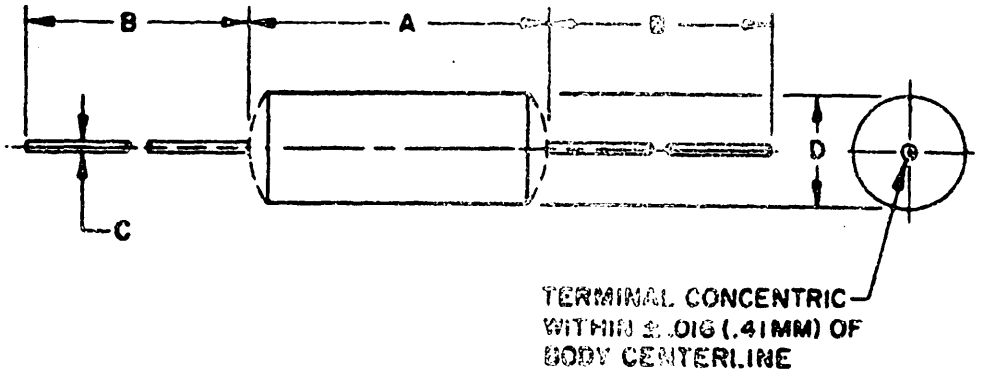
3.3 Minimum resistance value. The minimum resistance value available with each applicable tolerance shall be as follows:

Tolerance (± percent)	Symbol	Minimum resistance (ohms)
0.05	A	100
0.10	B	10
0.50	D	0.1
1.00	F	0.1

3.4 Maximum resistance value (based on use of 0.001-inch nominal diameter wire with a nominal resistivity of 800 ohms per circular mil foot). Maximum resistance value shall be 1.5 megohms.

3.5 Power rating. Power rating shall be 0.5 watt.

Source: https://assist.dla.mil -- Downloaded: 20261O-16T04:49Z
Check the source to verify that this is the current version before use.



Dim	Inches	Millimeters
A	1.000 ±.062	25.40 ±1.57
B	2.000 ±.125	50.80 ±3.18
C	.032 ±.002	.81 ± .05
D	.375 ±.031	9.53 ± .79

NOTE:
Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style RB52 resistor.

Source: https://assist.dla.mil -- Downloaded: 20261O-16T04:49Z
Check the source to verify that this is the current version before use.

3.6 Maximum voltage. Maximum voltage shall be 600 volts direct current or peak.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-93.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-93.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-93 are applicable to this specification.

6.2 Supplementary insulation. Where potential to ground is over 250 volts, supplementary insulation should be provided.

* 6.3 Substitution data. The resistors specified herein are not for use in design after date of this specification. They are authorized for use in design contracts effective prior to or on the date of this specification, and to support existing military equipment. Resistors specified in MIL-R-39005/1 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value and tolerance of this specification.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

(Project 5905-0660-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:49Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications - Electronics Command
ATTN: DRSEL-ED-TO
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:49Z
Check the source to verify that this is the current version before use.